

République Algérienne Démocratique et Populaire
Ministère de l'Enseignement Supérieur et de la Recherche Scientifique
ECOLE SUPÉRIEURE EN INFORMATIQUE
8 Mai 1945 - Sidi-Bel-Abbès



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وزارة التعليم العالي والبحث العلمي
المدرسة العليا للإعلام الآلي
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Introduction to Data Visualization

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1. Introduction

Currently, we are living in the era **of big data**. The volume of data used in businesses, industries, research organizations, and technological development **is massive**, and it is rapidly **growing every day**. The more data **we collect** and **analyze**, the more capable we can be in making **critical decisions**. However, with the enormous growth of data, it has become **harder to extract crucial information** from the available data. That is where the importance of data visualization becomes clear.



2. Data Visualization Fundamentals

2.1 Definition

Data visualization is the process of interpreting data and presenting it in a pictorial or graphical format.

Data visualization helps people understand the significance of data by summarizing and presenting a huge amount of data in a simple and easy-to-understand format in order to communicate the information clearly and effectively.



2.2 Why Is Data Visualization Important?

A picture is worth a thousand words, as they say. Humans just understand data better through pictures rather than by reading numbers in rows and columns. Accordingly, if the data is presented in a graphical format, people are more able to effectively **find correlations** and **raise important questions**.



- Data visualization will play a vital role, now and in the future, in applying many concepts such as **network theory**, **Internet of Things**, **complexity theory**, and more.
- ✓ For instance, network theory employs algorithms to understand and model pairwise relationships between objects to understand relationships and interactions in a variety of domains, such as crime prevention and disease management, social network analysis, biological network analysis, network optimization, and link analysis
- ✓ Internet of Things device data will provide extraordinary insight into what's happening around the globe.



- There are many ways in which visualizations help any field to improve its decision-making.

- ✓ **Faster Responses** : for instance, quick response to customers' or users' requirements is important for any company to retain their clients, as well as to keep their loyalty.
- ✓ **Simplicity** : It is impossible to make efficient decisions based on large amounts of raw data. Therefore, data visualization gives the full picture of the scoped parameters and simplifies the data by enabling decision-makers



- ✓ **Easier Pattern** : Visualization Data visualization provides easier approaches to identifying upcoming trends and patterns within data sets and hence enables businesses to make efficient decisions and prepare strategies in advance.
- ✓ **Team Involvement**: Data visualizations process not only historical data but also real-time data. Different organization units gain the benefit of having direct access to the extracted information displayed by data visualization tools.. **This increases the levels of collaboration between departments to help them achieve strategic goals**

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- ✓ **Unify Interpretation** :Data visualizations can produce charts and graphics that lead to the same interpretations by all who use the extracted information for decision-making



2-3 General Types of Visualizations:

Chart: Information presented in a tabular, graphical form with data displayed along two axes. Can be in the form of a graph, diagram, or map.

Table: A set of figures displayed in rows and columns.

Graph: A diagram of points, lines, segments, curves, or areas that represents certain variables in comparison to each other.



2-3 General Types of Visualizations:

Geospatial: A visualization that shows data in map form using different shapes and colors to show the relationship between pieces of data and specific locations.

Infographic: A combination of visuals and words that represent data. Usually uses charts or diagrams.

Dashboards: A collection of visualizations and data displayed in one place to help with analyzing and presenting data.



3. Overview of Visualization based tools

Data visualization tools help you to easily understand complex information by representing data through graphs or charts.

Data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data. Additionally, it provides an excellent way for employees or business owners to present data to non-technical audiences without confusion.

In the world of Big Data, data visualization tools and technologies are essential to analyze massive amounts of information and make data-driven decisions.

Yellowfin enables you to create stunning visualizations which can be used to build compelling dashboards, data stories and presentations, enabling everyone to see and understand trends, outliers, and patterns in their data.

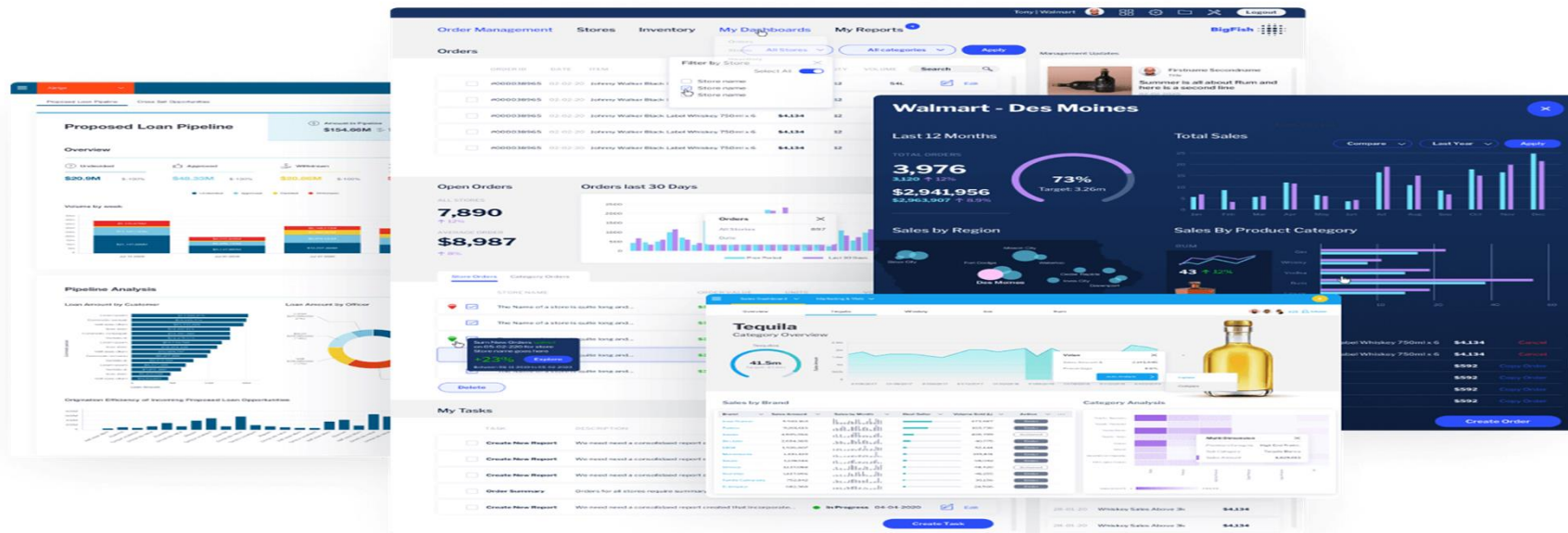


Figure 3.1: Yellowfin tool



Google Charts is an interactive Web service that creates graphical charts from user-supplied information. The user supplies data and a formatting specification expressed in JavaScript embedded in a Web page; in response the service sends an image of the chart.

The Google Chart gallery provides a large number of ready-to-use chart types:

Scatter Chart

Line Chart

Bar / Column Chart

Area Chart

Pie Chart

```

<!DOCTYPE html>
<html>
<script type="text/javascript" src="https://www.gstatic.com/charts/loader.js"></script>
<body>
<div id="myChart" style="width:100%; max-width:600px; height:500px;"></div>
<script>
google.charts.load('current',{packages:['corechart']});
google.charts.setOnLoadCallback(drawChart);
function drawChart() {
// Set Data
const data = google.visualization.arrayToDataTable([
  ['Price', 'Size'],
  [50,7],[60,8],[70,8],[80,9],[90,9],
  [100,9],[110,10],[120,11],
  [130,14],[140,14],[150,15]
]);
// Set Options
const options = {
  title: 'House Prices vs. Size',
  hAxis: {title: 'Square Meters'},
  vAxis: {title: 'Price in Millions'},
  legend: 'none'
};
// Draw
const chart = new google.visualization.LineChart(document.getElementById('myChart'));
chart.draw(data, options);
}
</script>
</body>
</html>

```

A Grafana dashboard consists of panels displaying data in beautiful graphs, charts, and other visualizations. These panels are created using components that transform raw data from a data source into visualizations.

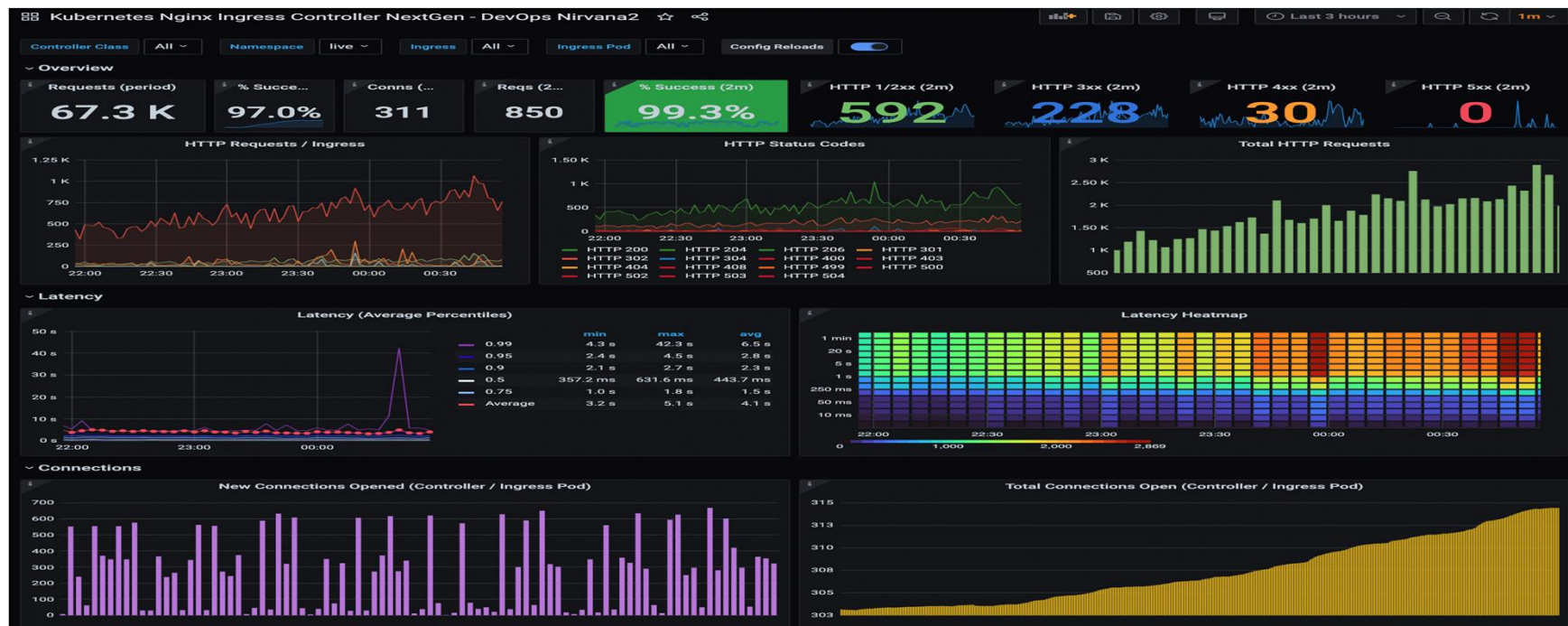


Figure 3.2: Grafana dashboard



4. Conclusion

Data visualization is the graphical representation of information and data. By using visual elements like charts, graphs, and maps.

Data visualization include:

- ✓ Easily sharing information.
- ✓ Interactively explore opportunities.
- ✓ Visualize patterns and relationships.